

LECTURE NOTES

Hard Problems in Cryptography

Mathematical Foundations for Post-Quantum Cryptography

Factorization · Discrete Logarithm · Lattices · Codes · Isogenies · Multivariate · Hash

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Classical hardness assumptions and post-quantum reductions

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Mathematical Foundations for Classical and Post-Quantum Cryptography

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Chapter 1

Preliminaries

1.1 Search vs. decision vs. distinguishing.

In post-quantum cryptography (PQC), many security assumptions are stated using one of three of hard problems:

1. **Search problem:** recover the hidden object.
2. **Decision problem:** answer a yes/no question about the instance.
3. **Distinguishing problem:** tell whether samples come from one distribution or another.

These three look similar, but they play different roles in theory and in cryptographic reductions.

Type	**Lattice**	**Code**
Search	SVP, CVP, BDD, SIS, Search-LWE	Syndrome Decoding, decoding, low-weight codeword search
Decision	GapSVP, GapCVP	existence of error of weight $\leq w$, gap decoding forms
Distinguishing	Decision-LWE, Ring-LWE, Module-LWE	structured code vs. random code, disguised secret code vs. random linear code

1.2 Preliminaries for Lattice-Based Hard Problems

This section collects the mathematical background needed to understand lattices, lattice basis reduction, and the hard problems that arise in post-quantum cryptography.

1.2.1 Notation

$\mathbb{Z}$ the ring of integers.

$\mathbb{R}$ the field of real numbers.

$\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$ integers modulo q.

$\mathbb{R}^n$ the n-dimensional real vector space.

$\mathbf{x}, \mathbf{y}, \mathbf{b}$ vectors.

A, B matrices.

$\langle \mathbf{x}, \mathbf{y} \rangle$ the standard inner product.

$\|\mathbf{x}\|$ the Euclidean norm of $\mathbf{x}$.

$\det(B)$ determinant of a square matrix B.

$\mathcal{L}(B)$ the lattice generated by the columns of B.

1.2.2 Vector spaces and Euclidean geometry

Vectors in $\mathbb{R}^n$

Definition 1.2.1. A vector in $\mathbb{R}^n$ is an ordered n-tuple

$$\mathbf{x} = (x_1, x_2, \ldots, x_n), \qquad x_i \in \mathbb{R}.$$

The set of all such vectors forms a vector space over $\mathbb{R}$.

Inner product

Definition 1.2.2. The standard inner product of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n.$$

This quantity measures the geometric alignment of two vectors.

Remark 1.2.3. Two vectors are said to be *orthogonal* if

$$\langle \mathbf{x}, \mathbf{y} \rangle = 0.$$

Euclidean norm

Definition 1.2.4. The Euclidean norm of a vector $\mathbf{x} \in \mathbb{R}^n$ is

$$\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Remark 1.2.5. This norm satisfies the standard properties:

1. $\|\mathbf{x}\| \geq 0$, with equality if and only if $\mathbf{x} = \mathbf{0}$,
2. $\|\alpha \mathbf{x}\| = |\alpha| \, \|\mathbf{x}\|$ for $\alpha \in \mathbb{R}$,
3. $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ (triangle inequality).

Linear independence and bases

Definition 1.2.6. A finite set of vectors

$$\{\mathbf{b}_1, \ldots, \mathbf{b}_m\} \subseteq \mathbb{R}^n$$

is *linearly independent* if

$$a_1 \mathbf{b}_1 + \cdots + a_m \mathbf{b}_m = \mathbf{0}$$

implies

$$a_1 = \cdots = a_m = 0.$$

Remark 1.2.7. If $\{\mathbf{b}_1, \ldots, \mathbf{b}_m\}$ is linearly independent, then its real span

$$V = \left\{ \sum_{i=1}^{m} a_i \mathbf{b}_i : a_i \in \mathbb{R} \right\}$$

is an m-dimensional subspace of $\mathbb{R}^n$, and the vectors $\mathbf{b}_1, \ldots, \mathbf{b}_m$ form a basis of V.

Remark 1.2.8. If B is the matrix whose columns are $\mathbf{b}_1, \ldots, \mathbf{b}_m$, then

$$V = \{B\mathbf{a} : \mathbf{a} \in \mathbb{R}^m\}.$$

Chapter 2

Lattice-Based Hard Problems

A (full-rank) lattice $\Lambda \subset \mathbb{R}^n$ is

$$\Lambda = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}$$

for some basis matrix $B \in \mathbb{R}^{n \times n}$. Classic algorithmic problems are

- **SVP (Shortest Vector Problem):** Find $\mathbf{0} \neq \mathbf{v} \in \Lambda$ minimizing $\|\mathbf{v}\|_2$.
- **GapSVP (Decision/SVP approximation):** Given (Λ, d) decide whether $\lambda_1(\Lambda) \leq d$ or $\lambda_1(\Lambda) > \gamma d$.
- **SIVP (Shortest Independent Vectors):** Find n linearly independent vectors of length $\leq \gamma \cdot \lambda_n(\Lambda)$.

The importance for cryptography: average-case problems (LWE/SIS) reduce from worst-case lattice problems (GapSVP/SIVP) under suitable parameters.

2.1 Learning With Errors (LWE)

TBA

2.2 Short Integer Solution (SIS)

TBA

2.3 NTRU Search Problem

TBA

Appendix A

Exercises for Lattice

A.1 Euclidean spaces and lattices

1. Let
$$\mathbf{x} = (1, -2, 3), \qquad \mathbf{y} = (4, 0, -1).$$
Compute $\langle \mathbf{x}, \mathbf{y} \rangle$, $\|\mathbf{x}\|$, $\|\mathbf{y}\|$, and $\|\mathbf{x} + \mathbf{y}\|$.

2. Determine whether the following sets are lattices in $\mathbb{R}^2$:
   (a) $\mathbb{Z}^2$
   (b) $\{(a, \sqrt{2}a) : a \in \mathbb{Z}\}$
   (c) $\{a(1, 0) + b(1/2, 1) : a, b \in \mathbb{Z}\}$
   (d) $\mathbb{Q}^2$

3. Let
$$\mathbf{b}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$
Describe explicitly the real span of $\mathbf{b}_1, \mathbf{b}_2$, and the lattice generated by them.

4. Show that if $\mathbf{b}_1, \ldots, \mathbf{b}_m \in \mathbb{R}^n$ are linearly independent, then
$$\mathcal{L} = \left\{ \sum_{i=1}^{m} z_i \mathbf{b}_i : z_i \in \mathbb{Z} \right\}$$
is a discrete subgroup of $\mathbb{R}^n$.

5. Give two distinct bases of the same lattice in $\mathbb{R}^2$.

6. Let $B = [\mathbf{b}_1 \ \mathbf{b}_2]$ with
$$\mathbf{b}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$
Write $\mathcal{L}(B)$ explicitly.

7. Show that a lattice may have infinitely many bases.

8. Let $\mathcal{L} \subseteq \mathbb{R}^n$ be a lattice. Prove that $\mathcal{L}$ is countable.

9. Show that every lattice contains the zero vector and is closed under addition and negation.

10. Let
$$\mathcal{L} = \{(2a + b, b) : a, b \in \mathbb{Z}\}.$$
Find a basis matrix B such that $\mathcal{L} = \mathcal{L}(B)$.

A.2 Gram–Schmidt orthogonalization

11. Apply the Gram–Schmidt process to
$$\mathbf{b}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

12. Apply the Gram–Schmidt process to
$$\mathbf{b}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{b}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{b}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

13. Let $\mathbf{b}_1, \mathbf{b}_2 \in \mathbb{R}^n$ be linearly independent. Show that
$$\mathbf{b}_2^* = \mathbf{b}_2 - \frac{\langle \mathbf{b}_2, \mathbf{b}_1 \rangle}{\langle \mathbf{b}_1, \mathbf{b}_1 \rangle} \mathbf{b}_1.$$

14. Prove that the Gram–Schmidt vectors $\mathbf{b}_1^*, \ldots, \mathbf{b}_m^*$ are pairwise orthogonal.

15. Prove that
$$\operatorname{span}_{\mathbb{R}}(\mathbf{b}_1, \ldots, \mathbf{b}_m) = \operatorname{span}_{\mathbb{R}}(\mathbf{b}_1^*, \ldots, \mathbf{b}_m^*).$$

16. Show by example that Gram–Schmidt does not in general preserve the lattice generated by the basis vectors.

17. Let
$$\mathbf{b}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$
Compute $\mu_{2,1}$ and $\mathbf{b}_2^*$.

18. Show that if the basis $\mathbf{b}_1, \ldots, \mathbf{b}_n$ is orthogonal, then $\mathbf{b}_i^* = \mathbf{b}_i$ for all i.

19. Let $\mathbf{b}_1, \ldots, \mathbf{b}_n$ be linearly independent. Show that
$$\mathbf{b}_i = \mathbf{b}_i^* + \sum_{j=1}^{i-1} \mu_{i,j} \mathbf{b}_j^*.$$

20. Explain why Gram–Schmidt is a geometric tool rather than a lattice basis reduction algorithm.